Screening Fusarium DNA and RNA reads for non-target sequences

Contamination detection practical | DNA and RNA | 20 October 2026

I examined one Fusarium DNA dataset (SRR27384773) and one RNA dataset (SRR1185280) to look for sequences from other organisms. I used both complete readsets without subsampling: 12,446,790 individual DNA reads and 27,608,126 individual RNA reads. Paired reads were counted as two individual reads in the mapping results.

First, I screened the reads with Kraken2 using the database already available on ASC, and with SendSketch using its online nt, RefSeq, and SILVA comparisons. These screens suggested candidates; a screen hit alone was not treated as proof. I then mapped reads against a combined reference panel containing the Fusarium host, a fungal virus, bacteria, and humans. Using the host and candidates together allowed competing matches to be considered. I kept primary mapped reads with mapping quality of at least 20 and calculated read percentage, average depth, and the percentage of reference bases covered.

The strongest finding was Fusarium graminearum virus 1 (FgV1) in the RNA dataset. A total of 14,497,528 reads, or 52.511815%, mapped to its 6,624-base reference. Coverage extended across 100% of the reference, with an average depth of approximately 220,615 times. This supports a biological detection worth reporting. It is also consequential for analysis because viral reads make up more than half the input RNA reads. A virus associated with the sampled fungal culture is more plausible than a few accidental matching reads, although these results alone cannot distinguish natural infection, experimental infection, or substantial sample carryover.

Human-reference matches were much smaller: 1,034 DNA reads (0.008307%) and 645 RNA reads (0.002336%). Their genome breadths were only 0.003989% and 0.001463%, respectively. Nine DNA contigs had human-best matches in the selected panel, making a trace human signal plausible. Handling or library carryover is a possible source, but shared sequences and reference ambiguity remain alternatives. I regard the DNA signal as provisional trace and the RNA signal as unresolved, conservatively classified as noise until specificity is checked. Three RNA reads matched Stenotrophomonas maltophilia; this is insufficient for a credible detection. Other tested bacterial references had no retained mapped reads.

I assembled DNA with MEGAHIT and RNA with rnaSPAdes, compared assembled sequences with the local reference panel, and made GC-versus-depth plots. Combining overlapping BLAST regions correctly changed 38 DNA contigs and 2,164 RNA transcripts from partial to broad query support. This correction did not change measured read depth. Remote BLAST was attempted but skipped because it was too slow. The main remaining limits are the restricted reference panel, capped BLAST alignments, and ambiguous mapping among similar RNA transcripts. The evidence supports a strong viral finding, while the small human and bacterial signals require more cautious interpretation.